

Introduction to Data Science Project

Course Booklet

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Course Requirements

Course Description

In this lab, you will experience a full Data Science lifecycle: starting with data preprocessing and ending with insights derived from visualizations and algorithmic analysis. Our objective is to showcase the various steps in the Data Science workflow so you can better understand the end-to-end process.

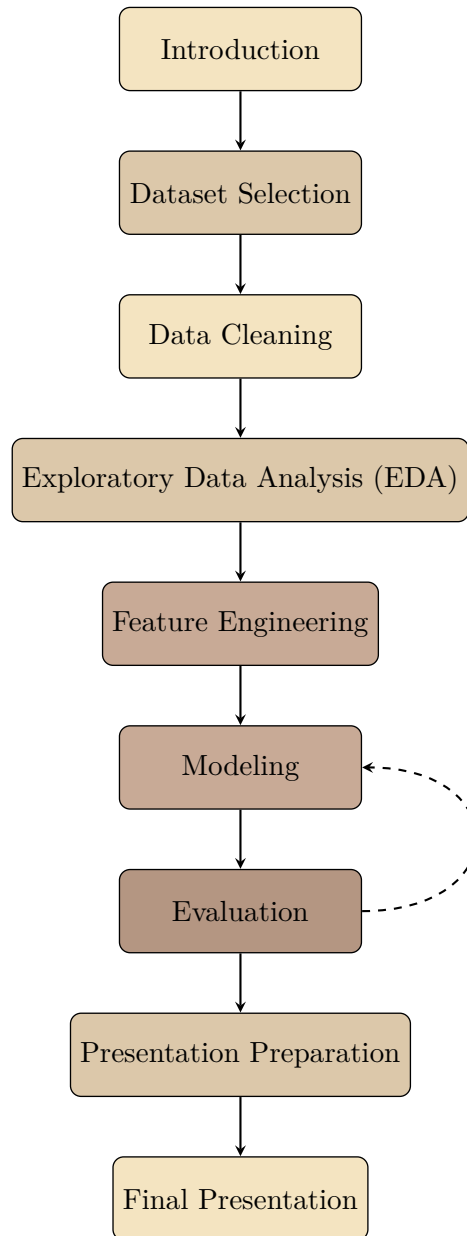


Figure 1: Project Workflow

Grading Breakdown

- ☐ Facilitator Evaluation - Meetings and Teamwork
- ☐ Final Project - report
- ☐ Presentations - middle and final

☐ GIT repo

☐ Performance

Datasets

IEMOCAP

Gannot Lab

The laboratory of Professor Sharon Gannot at Bar-Ilan University specializes in speech and audio signal processing, developing advanced algorithms for sound enhancement, noise reduction, and source localization. The lab's research integrates deep learning with statistical modeling to address complex acoustic challenges in telecommunications, hearing aids, and human-machine interfaces.

Data Introduction

The Interactive Emotional Dyadic Motion Capture (IEMOCAP) database is an acted, multi-modal and multispeaker database, recently collected at SAIL lab at USC. It contains approximately 12 hours of audiovisual data, including video, speech, motion capture of face, text transcriptions. It consists of dyadic sessions where actors perform improvisations or scripted scenarios, specifically selected to elicit emotional expressions. IEMOCAP database is annotated by multiple annotators into categorical labels, such as anger, happiness, sadness, neutrality, as well as dimensional labels such as valence, activation and dominance. The detailed motion capture information, the interactive setting to elicit authentic emotions, and the size of the database make this corpus a valuable addition to the existing databases in the community for the study and modeling of multimodal and expressive human communication.

link:

<https://sail.usc.edu/iemocap/index.html>

Gut Microbiome

Orr Levy Lab

The research lab is an interdisciplinary group dedicated to unraveling the complexities of biological network systems, with a particular focus on gene regulatory networks, aging, and the human microbiome. It develops innovative computational methodologies drawing on artificial intelligence, computational biology, ecology, nonlinear dynamics, statistical mechanics, and network science. Its theories and approaches are rigorously validated through carefully designed biological experiments conducted within the lab. The overarching goal is to advance the understanding of tissue biology and physiology, thereby contributing to improved diagnostics and therapeutics that can significantly enhance patient care. Central to its research is the investigation of the intricate relationships between the structure and dynamics of complex networks, which are fundamental to tissue function and stability.

Data Introduction

The gut microbiome, a vast community of microorganisms primarily located in our digestive system, plays a critical role in human health and is as unique as a fingerprint.

Research Goal- Quantify how exposure to “food microbiomes” affects key features of the human gut ecosystem, including stability, diversity, and overall community structure, to uncover mechanistic links between diet and gut health.

Link: <https://github.com/SegataLab/cFMD>,
https://github.com/knights-lab/dietstudy_analyses

Student Guide: Agile Development

PI Planning, User Stories, and the 8-Week Lifecycle

Welcome to the Project: Simulating Industry Work

The goal of this course is not just to write code, but to simulate a real-world tech environment as closely as possible. In the industry, no one works alone in a vacuum-work is done in teams, within strict timeframes (sprints), and using rigorous task and version management tools. Therefore, beyond your final product, **you will also be evaluated and graded on your workflow and teamwork.**

Project Structure and Competition

We will work on two main projects in this course. You have been divided into teams of 3-4 students. Every two teams will receive the exact same dataset and research question, working in parallel. The catch? You are actually competing against each other! Your grade will be determined by your performance and results compared to the parallel team.

Mandatory Tools: Git & GitHub Projects

Version Control (Git): Working with Git is **mandatory**. It is the most common industry tool for version control and code collaboration. *Note:* We will review and read your Git Commits! This is how we track individual contributions, ensure fair workload distribution, and assess your collaboration. For those who need to learn or brush up on this skill, here is a link to a 4-session learning series on the ZivTalks channel. It is your responsibility to learn this tool independently.

Task Management (GitHub Projects): To manage our development using Agile methodology, we will use **GitHub Projects**. Using this task board is an integral part of your grade. We will continuously monitor: who is assigned to which task, who is tracking their tasks, and who moves them to "Done". The initial task breakdown (PI Planning) is done together as a team, but within GitHub Projects, *you* decide which tasks actually enter each sprint and manage their progress.

Team Roles (Rotation)

To ensure everyone gains diverse experience, team roles will **rotate every two weeks** (at the start of each new sprint):

- **PO / Scrum Master:** Manages ongoing work. Responsible for ensuring the team meets its goals, that all tasks in GitHub Projects are updated and assigned, and helps resolve blockers.
- **Tech Lead:** Leads code quality. Responsible for Code Review (checking team members' code before merging to the main branch), making technical decisions, and ensuring proper architecture.
- **Data scientists:** The team members responsible for actually writing the code and developing features during that sprint (those who took on the tasks from the board).

Core Concepts

To work like a professional tech company, we follow Agile and SAFe (Scaled Agile Framework) methodologies. Here are the keys to your success:

- **PI (Program Increment):** A large timebox (usually 10-12 weeks in industry) where teams align on a common mission. In this course, your PI is **8 weeks**.

- **Sprint:** A short, fixed-length cycle (2 weeks) where you develop, test, and deliver a working piece of the project.
- **User Story:** The smallest unit of work. It describes a feature from the user's perspective: *"As a [User], I want to [Action] so that [Value]."*

Industry Ceremonies

Every sprint involves these key meetings to keep the team on track:

- **Daily Stand-up:** A 15-minute daily sync. What did I do? What will I do? Are there any blockers?
- **Sprint Planning:** Selecting stories from your backlog to work on during the current sprint.
- **Sprint Review/Demo:** Presenting your working software to stakeholders.
- **Sprint Retrospective:** Discussing what went well and what to improve for the next cycle.

The 8-Week Accelerated Schedule

Since our timeframe is condensed, we will run **4 Sprints**, each lasting 2 weeks. **Your task:** Fill in the Primary Focus and Expected Outcome for each sprint based on your team's PI Planning.

Phase	Primary Focus	Expected Outcome
PI Planning		
Sprint 1		
Sprint 2		
Sprint 3		

Success Tips

1. **Definition of Done (DoD):** Agree as a team on what "finished" looks like (e.g., code reviewed, tested, and merged).
2. **Stay Agile:** If a story is too big, break it down. If you're behind, reprioritize immediately.
3. **Documentation:** Update your GitHub Projects board daily. If it's not on the board, it doesn't exist.

ML Project – Meeting Preparation Checklist

Meeting Expectations

Teams are expected to make continuous progress throughout the semester, document their work clearly, and take responsibility for planning and executing their project.

Meetings are meant for discussion and guidance, not for first-time reporting of unfinished work. Teams should arrive prepared, with documented progress, results when relevant, and specific questions.

Because meetings occur only once every few weeks, teams are expected to make clear and measurable progress between meetings.

The project report should be *developed gradually during the semester*, rather than written at the end.

Before Each Meeting

1. Summary of the Previous Meeting

Write a short summary (5–10 lines):

- ☐ Main points discussed
- ☐ Team decisions and direction going forward

2. Progress Since the Last Meeting

Describe the work done since the previous meeting.

- ☐ Planned tasks
- ☐ What was completed / partially completed / not completed
- ☐ Who worked on what

3. Implementation / Code

Verify that:

- ☐ Code runs without errors
- ☐ Code is organized and commented in English
- ☐ Another team member can run the code

4. Results and Observations

Prepare material showing current progress.

- ☐ Experimental results, figures, or tables
- ☐ Key observations or insights from the results

5. Problems and Questions

List issues you would like to discuss.

For each issue include:

- Problem
- What you tried
- Question for discussion

6. Plan Until the Next Meeting

Propose the next steps.

- ☐ List next tasks
- ☐ Indicate who is responsible for each task

Report Progress

The project report should be **built gradually during the semester**.

- ☐ The report has been updated since the last meeting
- ☐ New results or figures are documented
- ☐ The method and experiments reflect the current system

Do not postpone writing the report until the end of the semester

Written Report Guidelines

Abstract

Briefly describe the problem addressed in the project, the approach used, and the main results obtained. The abstract should be about 5–7 sentences. This section is usually written near the end of the project.

Introduction

Describe the problem addressed in the project.

Include:

- the motivation for the problem
- why the problem is interesting or important
- the goals of the project

In addition, briefly describe existing approaches related to the problem.

Include:

- a few relevant papers, algorithms
- the main ideas of these approaches (in a few sentences)
- how they relate to your project

This section should provide context, not a detailed survey.

Methods

- dataset description
- data preprocessing
- Describe the algorithm or method used, including its mathematical formulation, explain how it is applied to the data and support your description with appropriate academic references to the relevant literature.

This section may evolve during the semester as the project develops.

Experiments and Results

Describe the experiments performed and present their outcomes.

For each experiment include:

- the goal of the experiment
- the setup used
- the results obtained

Include:

- tables of results
- figures or plots
- comparisons between approaches or parameters

All figures and tables should be explained in the text.

Discussion and Conclusion

Discuss and summarize the outcomes of the project.

Include:

- what worked well and what did not
- key insights from the experiments
- limitations of the approach
- possible improvements or future work

References

List references used in the project.

Presentation Guidelines

T.B.D